Simulation of an off-grid solar system to provide reliable energy access to the Island Community of Bellavista in Ecuador

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***Abstract*—This paper shows the design considerations of an off-grid solar power system working independently with an existing diesel generator to supply energy for lighting circuits and some low-power appliances installed in the Bellavista community. The main purpose of this design is to take advantage of the available infrastructure in the community to reduce initial investment costs while ensuring reliable energy access for the community. In addition, this system may be used as a starting point to build upon for future sustainable energy projects. The community is currently being powered through a diesel generator operating for 5 hours every day during the evening when the period of higher demand occurs. However, the community requires access to reliable energy access throughout the day to improve their quality of life. For the generator's analysis, we adapted its load profile from available data of previous studies carried out in the community. We used the HOMER Pro renewable energy software for the simulation and the Meteonorm 7.3 database to obtain the site's weather data file. The results show the sizing of the solar power system, its annual electricity production, and other indicators obtained from its operation. We also show the system's initial capital costs and the Net Present Cost, taking into account the existing equipment in the community. Besides, we estimate the annual electricity production of the generator, its amount of fuel used, its operating costs, and a baseline for greenhouse gas emissions produced in a year according to the actual operating conditions.**

***Keywords*—Reliable energy access, solar power system, HOMER Pro, diesel generator**

I. Introduction

According to the International Energy Agency (IEA), about 89% of the world's rural population had access to electricity in 2018, and it is expected that it continues to increase by about 4% more in the next decade [1]. This trend is similar in Central and South America, and by 2030, the projections of IEA indicate that almost 99% of the population would have access to electricity – about 2% more than in 2018. These trends show the committed efforts of the scientific community, governmental, and non-governmental entities to provide universal access to affordable, reliable, sustainable, and modern energy, as set by the targets from the United Nation's Sustainable Development Goals (SDG) 7 [2].

Several projects have been implemented in South America to supply the electricity needs of rural communities [3]–[5]. Most of the solutions are based on renewable energy systems, which are previously modeled using simulation tools to assess their feasibility and the initial investment costs [6]–[10]. Doctors and Kelman [11] used the HOMER Pro software to evaluate hybrid and solar power systems as alternatives to improve the electrification of the community of Porto Walter in the state of Acre, Brazil. Through this analysis, the authors determined the most energy-efficient and cost-effective system which was the solar photovoltaic alternative provided a lower Levelized Cost of Energy (LCOE) than the hybrid alternative. Similarly, Mamaghani et al. [12] compared the techno-economic viability of seven different renewable energy systems for rural electrification in Colombia through HOMER Pro simulations. The findings of this research show that the diesel-renewable configurations produced a very low carbon footprint and the high potential of the proposed hybrid combinations to cover the load demand of the studied communities.

A common problem affecting rural communities is the use of inefficient diesel generators, which provide a few hours of electricity supply and produce significant greenhouse gas (GHG) emissions. This is one of the issues that Ecuadorian rural communities face. Nevertheless, the research community's continued efforts have provided access to sustainable energy to remote communities in Ecuador. Urquizo et al. [13] presented an example of this through the rehabilitation of 20 solar home systems in the community of Cerrito de los Morreños, located in the Gulf of Guayaquil.

This paper presents the simulation of an off-grid solar power system using the HOMER Pro software. Bellavista, a community located on an island in the Gulf of Guayaquil,

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was selected as a case study for this paper. This system was designed to supply electricity to the lighting circuits and some low-power appliances (TVs, cell chargers, and DVD players) in the whole community, which works independently from the existing diesel generator in this site. The lighting circuits fed by the solar power system would work in parallel to the generator's lighting circuits. The main purpose of this design is to take advantage of the available infrastructure in the community to reduce initial investment costs while ensuring reliable energy access for the community. The availability of solar panels, a diesel generator, as well as a functioning distribution grid may reduce the initial investment costs for the system. The simulations and analyses are made for the first year of operation of the solar power system, therefore replacement costs of the equipment are not considered. Also, we estimate the annual electricity production of the generator, its amount of fuel used, and its greenhouse gas emissions produced in a year according to the actual operating conditions. This generator works 5 hours each day during the evening, which feeds the home appliances with higher power consumption and existing lighting circuits.

II. Methodology

The methodology of this study has been divided into different subsections to provide a clear description of the methodological sequence. It should be mentioned that sections II-D and II-E are the information required for the simulation of the scenario in Figure 2a. Likewise, sections II-F and II-G are the information required for the simulation of the scenario in Figure 2b.

A. Background of Bellavista Community

Bellavista is an island community located in the Province of Guayas, Ecuador. This community is located in the Gulf of Guayaquil approximately 28 miles from the city of Guayaquil (see Figure 1), which is the most important port city of the country.



Fig. 1. Geographic location of the community of Bellavista

The community has a population of 136 people, which includes 22 homes, a community center, a church, and a school without access to the national electricity grid [14]. The community has a diesel generator that works 5 hours per day all year round. The power of this generator is 45 kW and its hours of operation are from 18:00 to 23:00. During the rest of the day, the community has no electricity supply.

B. Climate

The community of Bellavista, according to the Köppen-Gaiger classification, a climate classification map, is included in the BSh group [15]. The places in this group are characterized by semi-arid and warm climates. Throughout Ecuador, there are two well-defined seasons, which are wet (January-April) and dry (May-December). The average monthly temperature ranges from 23.5^{o}C to 26.6^{o}C, with April being the warmest month. Likewise, the annual average humidity is 70%. The wind speeds in the area are very low, reaching an annual average value of 2 m/s. Also, the average annual global insolation is approximately 4.5 kWh/m^2. The meteorological data for this location were obtained from [16] [17].

C. Description of the parameters for the simulations

The goal of this simulation is to provide the sizing of the components for the solar system as well as a baseline analysis of GHG emissions from the existing generator in the community to provide reliable energy access to the community. The proposed scenario is to have the solar system supply the power for lighting and low-powered appliances during the day and have the diesel generator supply power for the rest of high-powered devices (refrigerators, washing machines, irons, and others) during its hours of operation, which initially are proposed to be the same as they currently are. A description of the proposed operation to be simulated in HOMER Pro for these independent power systems is provided in Figures 2a and 2b.

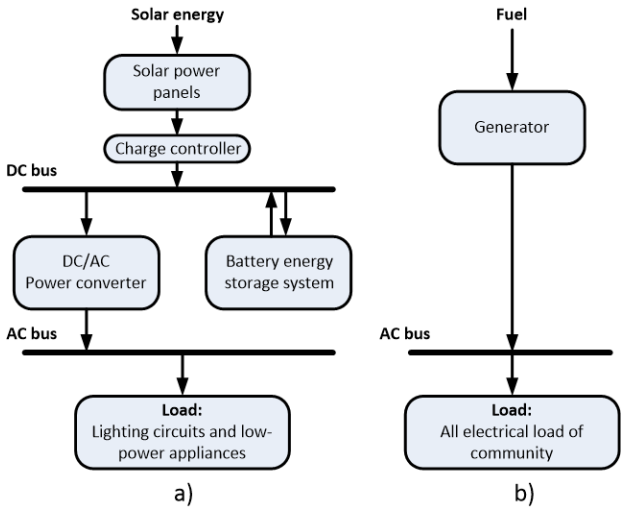


Fig. 2. Power Systems to be simulated in HOMER Pro a) off-grid solar power system, b) diesel generator

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D. Off-grid solar power system: Load profiles for homes

The load for a typical home in the community consists of lighting which is comprised mainly of Compact Fluorescent Light (CFL) bulbs, and appliances such as television, washing machine, cellphones. At least half of the homes also have a radio, DVD player, and a refrigerator. For design purposes, each home would have an electrical load and electricity consumption according to Table I. Each LED bulb circuit would work in parallel to the existing lighting circuits in each of the homes.

TABLE I
CHARACTERISTICS OF THE DAILY ELECTRICAL LOAD PER HOME

Appliance	Power (kW)	Qty. per home	Hours of operation	Energy (kWh)
LED bulb	0.009	4	6	0.216
Cell charger	0.005	1	1	0.005
TV	0.080	1	4	0.320
DVD player	0.030	1	3	0.090
			Total (kWh/day)	**0.631**

This paper considers the analysis for the 22 inhabited homes for the system as this information was readily available from previous assessments in the community. Taking into consideration the 22 homes, the daily electricity consumption for the entire community is 13.88 kWh/day, whose distribution can be seen in Figure 3. It should be mentioned that this distribution is referential. Likewise, the peak power reached was 2.69 kW with an average power of 0.58 kW. As can be seen, this load profile is only for when the solar power system is in operation.

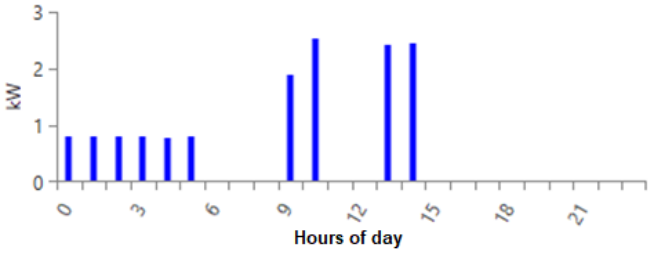


Fig. 3. Daily electricity consumption of 22 homes for electrical load estimation for the solar power system

The annual load profile of the whole community entered in the HOMER Pro was built taking into consideration a random daily variation of $\pm 10\%$ with an average daily electricity consumption of 13.88 kWh for the whole year.

E. Off-grid solar power system: Components

The selection of the components for the solar system was based on the availability of components on the community. Ten years ago the community received 22 Solar Home Systems (SHS) from an international non-governmental organization (NGO). Nowadays those SHS are no longer operational due to issues with maintenance of their batteries, as explored in [13]. However, the solar panels are still working and can be reallocated to a centralized system. For design purposes it is assumed that the community will donate their solar panels to connect them in a centralized system for the benefit of the community.

1) Solar power panels: The panel selected for the simulation has an efficiency of 15% in standard conditions, it is mono-crystalline and it has a lifetime of 15 years. The costs shown in Table II, which are input requirements for the analysis on HOMER pro, are based on referential mono-crystalline solar panels of similar peak power that are locally available.

TABLE II
COSTS OF SOLAR POWER PANEL

Capacity (kW)	Capital (USD$)	Replacement (USD$)	Maintenance (USD$)
0.155	150.00	0.00	0.00

In the simulator, the temperature effects on the power and the nominal operating cell temperature in the solar panel were taken into consideration. These values are -0.450 %/^{o}C and 46°C, respectively. Similarly, the derating factor is 98%, where 2% represents the losses such as dust on the surface of the panels, wiring, shading, and others. The nominal panel voltage is 17.3V with an operating current of 8.96A. The stacking of 4 panels makes it suitable for use with systems of 48V.

2) Battery energy storage system: In the present work, we take into account the stationary 2V - 1990 Ah OPzS Solar batteries, due to their higher capacity and lifespan. The batteries have a capital cost of USD $842 per unit with a maximum depth of discharge of 20%, and the replacement costs are not considered. Similarly, we do not consider the operation and maintenance costs for this first year. The specifications of the battery to be used can be found in Table III.

TABLE III
SPECIFICATIONS OF 1990 AH SOLAR OPZS BATTERY, [18]

Nominal voltage (V)	Maximum charge current (A)	Maximum discharge current (A)	Performance HOMER pro (kWh)	No. of batteries in series
2	203	8900	7300	24

According to the number of panels obtained in the simulation and their array type, the characteristics and costs of the solar charge controller will be established.

3) DC/AC power converter: The costs of the equipment used for optimization in the HOMER Pro can be seen in the Table IV.

TABLE IV
COSTS OF DC/AC POWER CONVERTER

Capacity (kW)	Capital (USD$)	Replacement (USD$)	Maintenance (USD$)
10	7556.00	0.00	0.00

This equipment has an operating voltage of 48 V and an efficiency of 96% [19]. Its lifespan is 15 years for which replacement costs are not considered. It has a low failure rate so it has no operation and maintenance costs.

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F. Diesel generator: Load profiles for homes

The existing electrical load to be supplied by the generator in the community is summarized in Table V. It was constructed from the information obtained in [14]. It should be specified that the generator would continue to work for 5 hours from 18:00 every day of the year for the proposed system.

TABLE V
CHARACTERISTICS OF THE DAILY ELECTRIC LOAD BY ALL HOMES

Appliance	Power (kW)	Qty. for all homes	Hours of operation	Energy (kWh)
Computer	0.065	1	3	0.195
Saving bulb (3 units)	0.075	22	5	8.250
Washing machine	0.400	18	1	7.200
Cell charger	0.015	15	1	0.225
Iron	1.200	1	1	1.200
Refrigerator	0.065	8	5	2.600
TV	0.095	22	5	10.450
DVD player	0.010	11	3	0.330
			Total (kWh/day)	**30.45**

The daily electricity consumption of the 22 homes is 30.45 kWh/day, whose distribution can be seen in Figure 4. Likewise, the peak power reached was 12.92 kW with an average power of 1.28 kW.

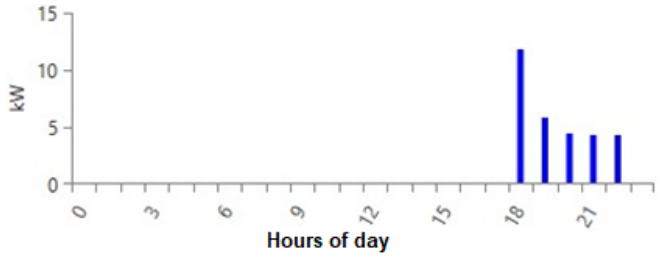


Fig. 4. Daily electricity consumption of 22 homes for electrical load estimation for the diesel generator

The construction of the annual load profile follows the same methodology as the one mentioned in Section II-D.

G. Diesel generator: Characteristics of the fuel and generator

The present information is valid for the simulations taking as reference Figure 2b. The fuel used by the generator is diesel, and its properties were extracted from fuel quality research (Table VI) [20]. Since the analysis is performed for one year, fuel price inflation is not considered. The information from the generator will only be used to estimate GHG emissions.

TABLE VI
PROPERTIES OF DIESEL FUEL

Fuel	Price ($/L)	Lower heating Value (MJ/kg)	Density (kg/m3)	Carbon content (%)	Sulfur content (%)
Diesel	0.274	43.957	880	81.208	0.184

The existing generator in the community has an output power of 45 kW. The fuel consumption curve in Figure 5a relates the working power to the amount of fuel per hour. This allows us to determine the efficiency curve of the generator (Figure 5b). These curves were obtained from the manufacturer data sheet. [21].

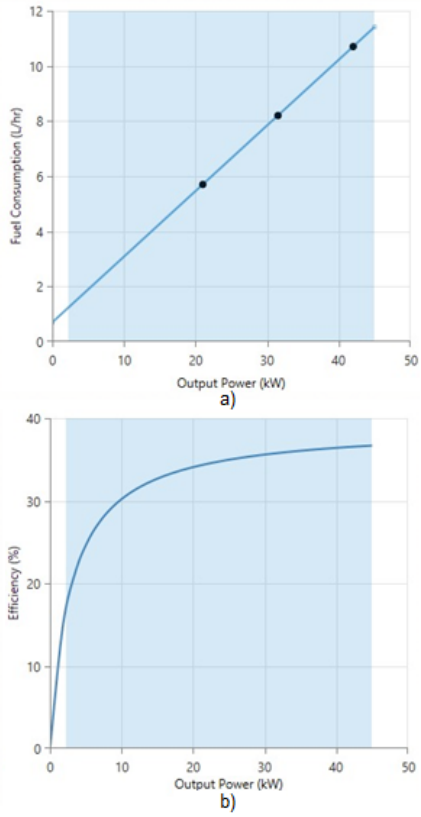


Fig. 5. Characteristics of the diesel generator obtained by HOMER Pro a) Fuel consumption b) Efficiency

For the system costs, the estimated capital cost of the generator is considered to be USD$0, because this machine is already deployed in the community. The replacement costs are not considered because the analysis is considered for a single year. Besides, the cost of operation and maintenance (O&M) per hour considered is USD$0.45.

III. RESULTS

This section shows the results of the two simulated power systems in HOMER Pro established in Section II-C. The first part is based on the simulations in Figure 2a and the second part in Figure 2b.

A. Off-grid solar power system

The simulations carried out are based on Figure 2a taking into consideration the information of sections II-D and II-E. According to the optimization analysis developed in HOMER Pro, the solar system was sized to a 4.96 kWp solar array. Each panel is 0.155 kW, so we have an array of 4 panels in series and 8 strings in parallel (32 panels) to work with an operating voltage of 48V. During a visit in January 2019, a group of IEEE volunteers tested the solar panels in the community, and 19 were found to be in operating conditions.

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With this arrangement and with the short circuit current of the selected panel (9.18A), we obtained an approximate current of 79A. This value should be increased by an additional 25% as a safety factor. So, we need a solar charge controller that supports at least 99A and it is suitable for the operating voltage of the system.

The solar power delivered by the system can be seen in Figure 6. The annual electricity production of the system reached 6.5 MWh/yr. Likewise, the electrical load consumed 5.11 MWh/yr. Only, 1.90 kWh/yr was the unmet electric load with excess electricity of 0.7 MWh/yr. Solar power panels operated for 4406 hours/yr with a maximum power of 3.82 kW and an average output of 0.74 kW. The average amount of energy supplied daily was 17.8 kWh/day.

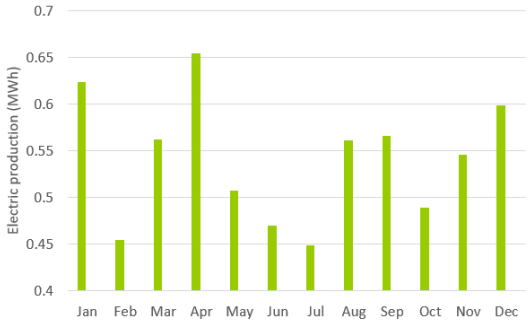


Fig. 6. Monthly electric production of solar power system

With the selected batteries, the system autonomy reaches 145 hours. The energy input of the batteries was 3.098 MWh/yr and the output was 2.64 MWh/yr. The losses reached by the storage system were 0.47 MWh/yr. According to the optimization made in HOMER Pro, the number of batteries is 24, which are connected in series. Each battery is 2V, so the operating voltage of the system is 48V.

The power converter had a minimum capacity of 4.13kW with an average output of 0.583 kW and a maximum output of 2.69 kW. The operation hours of this equipment during the year were 3650. Likewise, its input power was 5.3 MWh/yr and the output power was 5.11 MWh/yr, where a 96% efficiency of the equipment is clearly indicated. A summary of the energy produced and consumed for all of the components is shown in Fig. 7.

The capital costs of the components can be seen in Table VII. The estimated costs of these components come from the Ecuadorian market. Here, we are also considering the cost of 13 additional solar power panels of the 32 needed for the system, because in the community there are 19 panels in useful conditions. In "Others" we considered the costs of the solar charge controller [22] and the shipping/assembly of the components. The total cost of USD$33758 represents the Net Present Cost (NPC) of the solar power system because the system is being analyzed in its first year of operation.

B. Diesel generator

In this section, the simulations are based on the Figure 2b, taking into consideration the information of sections II-F and

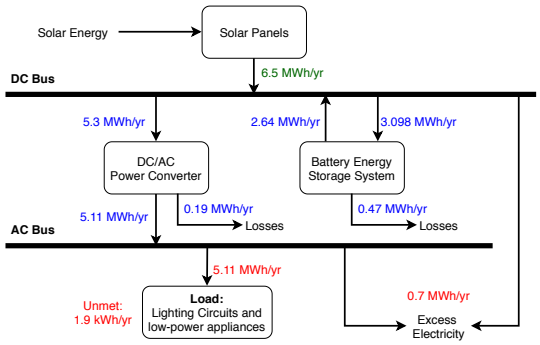


Fig. 7. Summary of the energy produced and consumed for all of the components.

TABLE VII
CAPITAL COSTS OF COMPONENTS OF SOLAR POWER SYSTEM

Components	Unit cost (USD$)	Quantity	Total cost (USD$)
Solar power panel	150	13*	1950
Battery	842	24	20208
Power inverter	3080	1	3080
Others	8520	1	8520
		Total (USD$)	33758

*During a visit in Jan/2019, 19 solar panels were tested.

II-G. As mentioned in the previous sections, the community has had a diesel generator for several years, which was donated by an Ecuadorian government entity. The diesel generator supplied 11.24 MW/yr, which represents 100% of the community's electrical load within its operating hours. The monthly power delivered by the generator can be seen in Figure 8.

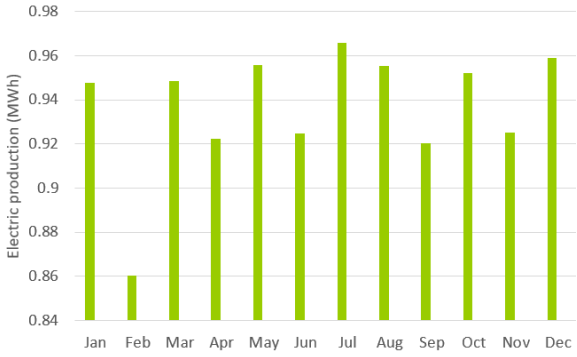


Fig. 8. Monthly electric production of diesel generator

The hours of operation of the generator were 1825 with 365 starts during one year of operation. The mean electrical output power was 6.16 kW with a minimum of 3.87 kW and a maximum of 12.9 kW.

The total annual diesel consumption was 3953 liters with a daily average of 10.8 L/day and per hour of 0.451 L/hr. Also, the monthly average diesel consumption was 329 L/month. The total cost of fuel during the year of the generator's operation was USD$1023.03. The NPC for the operation of the generator in one year was USD$1799. It is because the generator exists in the community and only the costs of

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O&M (USD$775.63) and fuel (USD$1023.03) are taken into account. Finally, the calculation of the pollutant gases, emitted by the generator from the information in Table VI, can be seen in Table VIII.

TABLE VIII
GREEN HOUSE GASES

Gases	Emission (kg/year)
Carbon dioxide	45429
Carbon monoxide	344
Unburned hydrocarbons	12.5
Sulfur dioxide	111
Nitrogen oxide	391

IV. CONCLUSIONS

This paper shows the design considerations of an off-grid solar power system that supplies energy to an electrical load consisting of lighting circuits and some existing low-power appliances in all the homes in the Bellavista community. The design of the system considers the operation of the existing diesel generator to improve energy access to the community, taking into consideration a low initial investment cost.

The availability of electricity for lighting would be of great help to the inhabitants of this community as they are fishermen and they start their work very early in the morning. Also, artificial lighting could help to improve the visual comfort level of the people so they could improve the development of other activities such as reading and domestic chores, during the daytime.

Sizing of system components resulted in values suitable for a 48 V system. We obtained 24 batteries and 32 solar panels, which indicates that the inputs entered into the simulator are consistent according to the assumptions initially established.

The combustion that takes place in the diesel generator produces large amounts of GHGs. It can directly affect the inhabitants' health as well as the biodiversity and seawater bodies. This situation could have a negative impact on the main economic activity of the inhabitants of this community, which is fishing and crustacean gathering.

This work provides a baseline for GHG emissions from the generator which may be reduced in the future as a greater renewable energy penetration.

Prior to the implementation of the system, community engagement must be prioritized in order for the community to have ownership of the project. This will allow the project to be sustained in time as the community becomes more involved in the implementation and maintenance of the system. Training would also be required for the maintenance of the different components as well as to keep the community educated on the scalability of solar systems and managing expectations for increasing the load of the system.

REFERENCES

[1] IEA, "SDG7: Data and Projections," Paris, 2019, [Online]. Available: https://www.iea.org/reports/sdg7-data-and-projections.

[2] United Nations, "Sustainable Development Goals: 7 Affordable and Clean Energy," 2020, [Online]. Available: https://www.un.org/sustainabledevelopment/energy/.

[3] M. Bueno-López and S. Garzón Lemos, "Electrification in non-interconnected areas," *IEEE Technology and Society Magazine*, vol. 36, no. 4, pp. 73–79, 2017.

[4] M. Jensen, E. Hopps, and B. Roth, "Assessing the Impact of Off-grid Solar Electrification in Rural Peru," *International Journal for Service Learning in Engineering, Humanitarian Engineering and Social Entrepreneurship*, vol. 14, no. 3, pp. 61–79, 2019.

[5] T. T. Sepulveda and L. Martinez, "Optimization of a hybrid energy system for an isolated community in Brazil," *Int. J. Renew. Energy Res. IJRER*, vol. 6, no. 4, pp. 1476–1481, 2016.

[6] A. López-González, B. Domenech, and L. Ferrer-Martí, "Formative evaluation of sustainability in rural electrification programs from a management perspective: A case study from Venezuela," *Renewable and Sustainable Energy Reviews*, vol. 95, pp. 95–109, 2018.

[7] A. López-González, B. Domenech, D. Gómez-Hernández, and L. Ferrer-Martí, "Renewable microgrid projects for autonomous small-scale electrification in Andean countries," *Renewable and Sustainable Energy Reviews*, vol. 79, pp. 1255–1265, 2017.

[8] A. Vides-Prado, E. O. Camargo, C. Vides-Prado, I. H. Orozco, F. Chenlo, J. E. Candelo, and A. B. Sarmiento, "Techno-economic feasibility analysis of photovoltaic systems in remote areas for indigenous communities in the Colombian Guajira," *Renewable and Sustainable Energy Reviews*, vol. 82, pp. 4245–4255, 2018.

[9] J. Barzola, M. Espinoza, and F. Cabrera, "Analysis of hybrid solar/wind/diesel renewable energy system for off-grid rural electrification," *Int. J. Renew. Energy Res*, vol. 6, no. 3, pp. 1146–1152, 2016.

[10] R. Hidalgo, P. Jacome, J. Urquizo, and G. Soriano, "Microgrid design for remote location in chile using a scenario-based methodology," in *14th LACCEI International Multi-Conference for Engineering, Education, and Technology:"Engineering Innovations for Global Sustainability*, 2016, pp. 20–22.

[11] R. Doctors and R. Kelman, "Sustainable and Economic Alternative for the Electric Supply of Isolated Communities in the Brazilian Off-Grid Systems," *Journal of Power and Energy Engineering*, vol. 5, no. 09, p. 145, 2017.

[12] A. H. Mamaghani, S. A. A. Escandon, B. Najafi, A. Shirazi, and F. Rinaldi, "Techno-economic feasibility of photovoltaic, wind, diesel and hybrid electrification systems for off-grid rural electrification in Colombia," *Renewable Energy*, vol. 97, pp. 293–305, 2016.

[13] J. Urquizo, P. Singh, R. Hidalgo-León, V. Villavicencio, and G. Soriano, "Rehabilitation of Solar Home Systems and Sustainable Development for an Island Community in Ecuador," in *2019 IEEE Global Humanitarian Technology Conference (GHTC)*. IEEE, 2019, pp. 1–8.

[14] G. M. Panchana Malavé, R. E. Valverde Rea *et al.*, "Diseño del sistema de energización autónomo de la isla bellavista localizada en el golfo de guayaquil," B.S. thesis, Espol, 2019.

[15] M. C. Peel, B. L. Finlayson, and T. A. McMahon, "Updated world map of the köppen-geiger climate classification," 2007.

[16] Meteonorm V.7.3, "Weather data report for the Bellavista community." ESPOL, Guayaquil, Tech. Rep., 2020.

[17] D. Vaca and F. Ordóñez, "Mapa Solar del Ecuador," Scinergy, Tech. Rep., 2018. [Online]. Available: https://www.researchgate.net/publication/338843581

[18] Power GNB Industrial, "Industrial Batteries / Network Power." [Online]. Available: http://www2.exide.com/Media/files/Downloads/IndustEuro/Classic Solar.pdf

[19] Victron Energy, "Inversores / Cargadores." [Online]. Available: https://www.victronenergy.com.es/upload/documents/Datasheet-Quattro-3-10kVA-120V-ES.pdf

[20] S. H. Medina Romo, J. Viteri Moya, R. Salazar Reyes, and G. L. Carvajal Chávez, "Composición elemental de mezclas biodiésel-diésel y análisis de sus emisiones de combustión," 2017.

[21] Sivasa-ec, "Grupo electrogeno MP-45." [Online]. Available: http://www.sivasa-ec.com/fichas/812.pdf

[22] Steca Power Tarom, "Technical data." [Online]. Available: https://www.steca.com/index.php?Steca-Power-Tarom-en#productproperty

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